

CERTIFICATE

This is to certify that Mr./ Ms. _____ Of
Division_____, Branch_____, Batch_____ Roll
No._____, has satisfactorily completed his / her term work in the
subject **MECHANICS OF SOLIDS (CL243)**for the term ending in
_____20___ / 20___.

Date:

Sign of the Faculty

Head of the Department

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

INDEX

Sr. No.	Title	Page No.	Date	Grade	Signature
1.	Izod Impact Test				
2.	Rockwell Hardness Test				
3.	Transverse Test on Timber				
4.	Compression Test on Bricks & Blocks				
5.	Tension Test on Various Materials				
6	Tutorial:1 Simple Stresses and Strains				
7	Tutorial:2 Shear Force and Bending Moment				
8	Tutorial:3 Moment of Inertia				
9	Tutorial:4 Principal Stresses and Strains				
10	Tutorial:5 Bending and Shear Stresses in Homogeneous and Composite Beam Sections				
11	Tutorial:6 Strain Energy				

Experiment No. 1 IZOD IMPACT TEST

OBJECTIVE:

To study the behavior of metals under impact load.

EQUIPMENT:

Izod Impact Testing Machine.

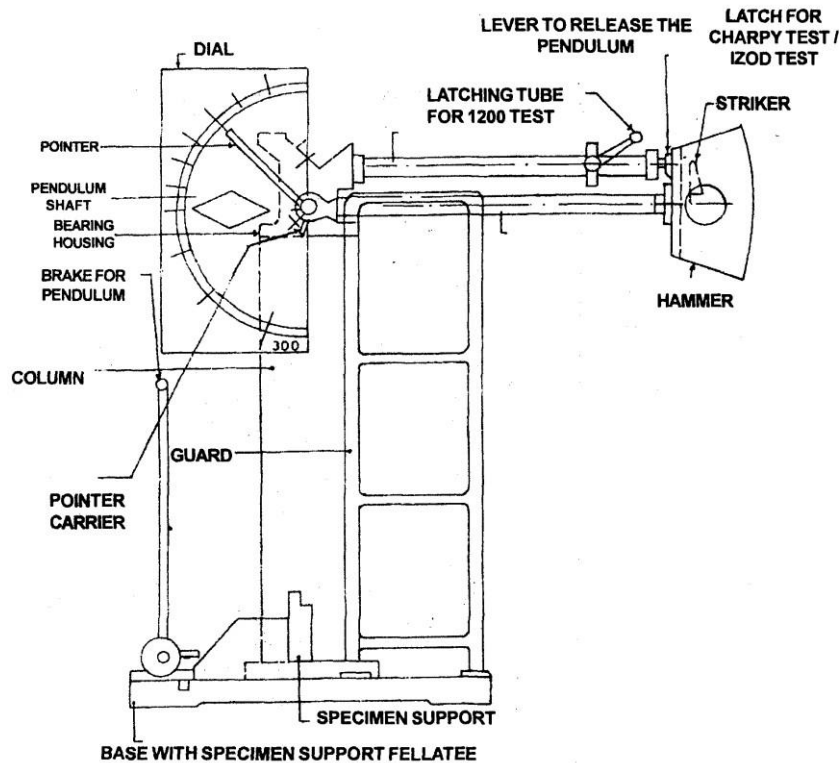


Fig. 1.1 Izod Impact Test Machine

SPECIMEN:

Mild steel, Aluminum and Brass.

RELATED I.S. CODE:

IS: 1598 – 1997, Methods for Izod impact test on metals.

THEORETICAL BACKGROUND:

- **Izod Impact Test:** Impact loads are applied suddenly and this test provides information about the behavior. In this test, the specified cantilevering length of the specimen is projecting out from a vise with a notch at the base, and a swinging pendulum strikes on the edge of the specimen giving a flexural type of loading of materials under such loads.

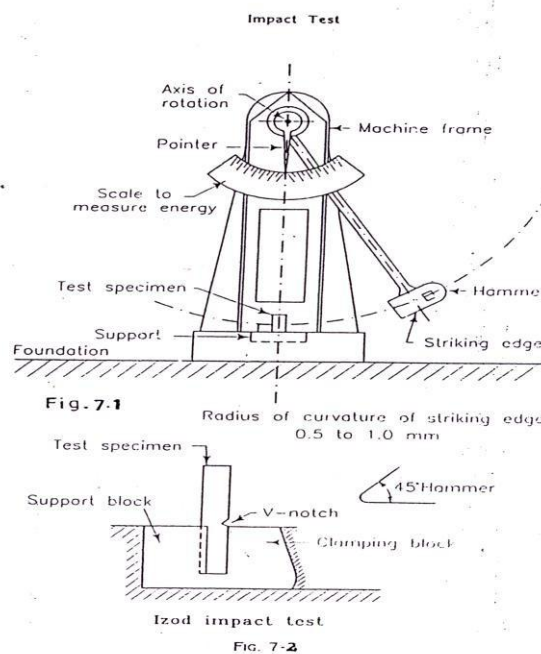


Fig. 1.2 Izod Impact Test on Specimen

- **Impact Resistance:** Impact resistance is the capacity of material to absorb energy, which depends on toughness of material.
- **Toughness:** Energy required to rupture a material.

SPECIMEN:

Specimen with notch at suitable position is used for carrying out the impact test on metals. For steel specimen, requirement is as follow:

Shape of The Specimen	Type	Type of Notch	Reference
Square or Round	Cantilever beam fixed at notch	V notch at base	Fig. 1.3

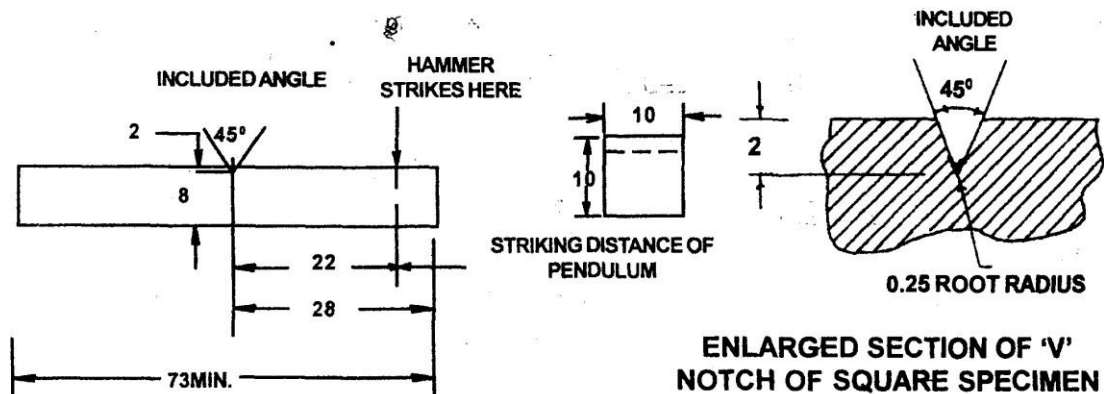


Fig. 1.3 Single Notch Square Specimen for Izod Impact Test Confirming to I.S. 1958 – 1960

PROCEDURE:

1. Check the specimen, measure its dimensions and ascertain the IS requirements are satisfied.
2. Lift the hammer in position. Put the stop bar and set the indicator on the scale.
3. Fix the specimen between the jaws properly such that the notch is in level with the anvil and faces the hammer side and the level of the specimen is in line with the striking edge of the hammer.
4. Release the pendulum and allow it to strike the test specimen.
5. Note the reading on the scale indicate by the pointer after the specimen is fractured, which gives impact value.
6. Study the type of fracture and co-relate it with fracture of specimens of different material.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

PRECAUTIONS:

1. It is not advisable to stand near or in front of the machine when the pendulum is to released.
2. Notch should be exactly in the line of action of the pendulum.
3. When specimen is being fitted, care should be taken that pendulum dose not get released.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

APPLICATIONS:

1. This test is basically designed so as to check the suitability of materials subjected to dynamic loads, e.g., vehicles traveling over bridges, hull of ship subjected to waves or hammer falling on nail.
2. The ductile material like steel has more impact value and therefore has wide acceptability in structural and other mechanical application.

OBSERVATION TABLE:

Sr. No.	Material	Energy Losses (J)	Impact Value (N m)	Mode of Failure
1.	Mild Steel			
2.	Aluminum			
3.	Brass			

OR					
Sr. No.	Material	Gross Energy (J)	Energy Losses (J)	Impact Value (N m)	Mode of Failure
1.	Mild Steel				
2.	Aluminum				
3.	Brass				

CONCLUSIONS:

Grade Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

Date:

EXPERIMENT NO. 2 ROCKWELL HARDNESS TEST

OBJECTIVE:

To determine the hardness number of given specimens by Rockwell hardness test.

EQUIPMENT:

Rockwell Hardness Testing machine with a direct reading dial.

APPARATUS:

Microscope.

MATERIALS:

Mild steel, Brass and Aluminium.

RELATED I.S. CODE:

IS: 1586 – 1986: Method of Rockwell hardness test for metallic materials.

FIGURE:

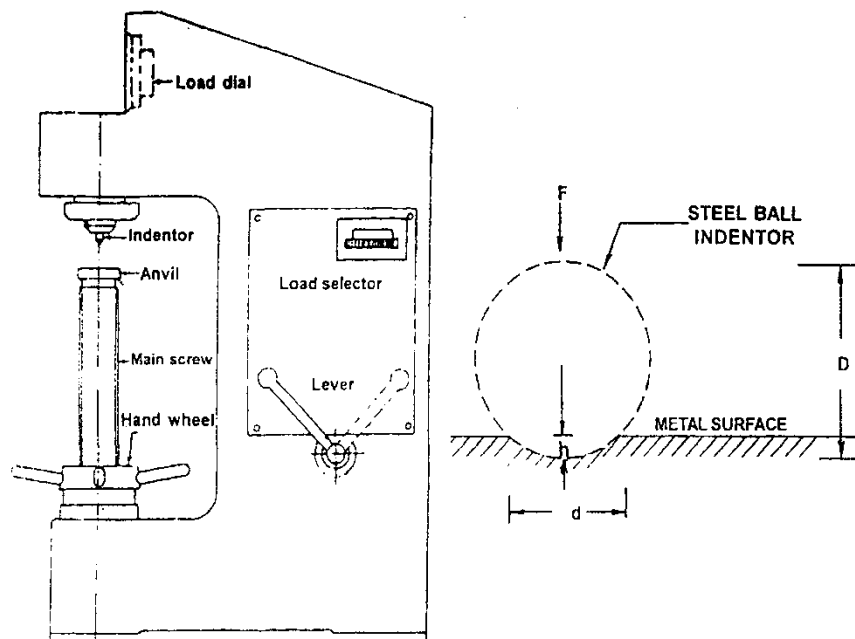


Fig. 2.1 Rockwell Hardness Tester

THEORETICAL BACKGROUND:

Hardness is an important property of metals. It is defined as a resistance given by the metals specimen to scratching, abrasion or indentation on its surface.

This experiment performs to indentation hardness under a given static load. Standard value of the indenting force is applied on the specimen and impression formed on the surface of the specimen is measured is assigned. The prefix HR with dial reading is used to designate the Rockwell hardness number. The line diagram of the machine is shown in the **Fig. 2.1**.

Rockwell number is directly read from dial having B& C scales. B-scales are meant for materials with of medium hardness whereas C-scale is used for harder than HRB 100. The C-scale should not be used for range below HRC20.

This test differs from the Brinell test in the sense that the penetration and the applied loads are smaller, and hence the resulting indentation is shallower, so it is applicable to the testing of materials having hardness beyond the scope of the Brinell test. Rockwell hardness number is inversely proportional to the depth of the indentation. Fig. 2.1 shows the indentation caused by steel ball and the indentation caused by diamond cone on the metal surfaces. A minor load of 10 kg is applied first for proper fixing of specimen, which causes initial indentation and sets the indenter on the specimen. This is followed by the application of major load, which leaves an indentation on the surfaces of the specimen. The depth of the indentation (h) is derived from the dimension of the indentation.

$$\text{Rockwell B number (HRB)} = 130 - \text{depth of penetration } h \text{ in mm} / 0.002$$

$$\text{Rockwell C number (HRC)} = 100 - \text{depth of penetration } h \text{ in mm} / 0.002$$

Calculation of depth of penetration:

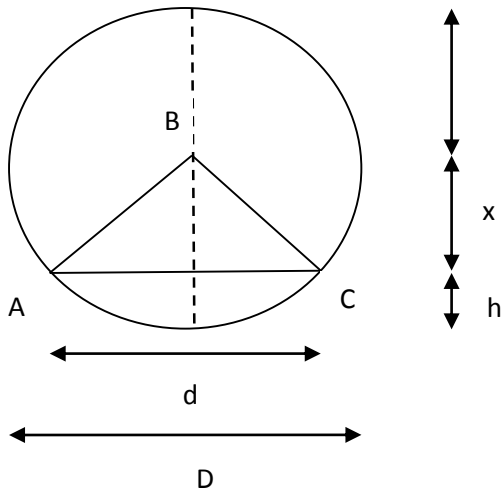


Fig. 2.2

From figure 1.2

$$x = (D/2) - h$$

$$h = \frac{[D - \sqrt{D^2 - d^2}]}{2}$$

Where D is dia of indenter and

The tabulated details of the load to apply and scale to be referred for a particular material are given in **Table 2.1**.

Table 2.1 Loads Applied and Scale for Different Materials

Scale	Indenter Type	Major Load (Kg)	Dial	Typical Applications of Scales
B	Hardened steel Ball of dia. 1.5875mm (1/16" ball indenter)	100	Red	Copper alloys, aluminum alloys, soft steel etc.
C	Diamond cone angle at tip (120±0.5) (tip of diamond cone is rounded to radius of 0.2 mm) (diamond ball indenter)	150	Black	Steel, Hard C.I deep case hardened steel etc.

PROCEDURE:

1. Place the specimen on the anvil.
2. Apply minor load of 10 kg gradually to ensure proper holding of load to specimen.
3. Select proper value of load with the help of load selector.
4. Adjust the pointer at "set" position and set the dial to zero position.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

5. Apply major load gradually by operating handle without any interference.
6. Bring the lever back to its catch position to take off the load from the specimen.
7. Read position of the pointer on appropriate dial that gives Rockwell hardness number.
8. Measure the diameter of the impression with the help of microscope and determine the depth of the indentation using simple geometry.

PRECAUTIONS:

The minimum distance between two indentations should be at least $2d$ and the distance of the indentation from edge of the specimen should be at least $3d$, where d is the diameter of the indentation.

APPLICATIONS:

On the basis of hardness, materials are graded for their commercial use. The quality of materials and its products is maintained or controlled by hardness test. Also the strength of the job like forging, alloying case hardening etc is determined with the help of this test.

Some correlation of hardness with other parameters like tensile strength is established which is useful in determining the tensile strength of the material.

OBSERVATION TABLE:

Sr. No	Specimen	Scale	Indenter Type	Major Load (kgf)	Dial Colour	Rockwell hardness Number	Diameter of Indentation 'd' (mm)
1.	Mild Steel						
2.	Brass						
3.	Aluminum						

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CALCULATIONS:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

RESULT TABLE:

Sr. No.	Specimen	Hardness Number		Remarks
		Experimental	Analytical	
1.	Mild steel			
2.	Aluminium			
3.	Brass			

CONCLUSION:

Grade Obtained:	Signature of Faculty:	Date:
------------------------	------------------------------	--------------

Date:

EXPERIMENT NO. 3 TRANSVERSE TEST ON TIMBER

OBJECTIVE:

To study the behavior of a timber specimen under transverse or flexural loading and determine the bending properties such as Bending stress, Young's modulus of elasticity and Modulus of rupture.

EQUIPMENT:

Universal Testing Machine.

APPARATUS:

Vernier-calliper.

MATERIAL:

Timber.

RELATED I.S. CODE:

IS: 1708 – 1986 (Part 8 & 9): Methods of testing of small clear specimens of timber.

FIGURE:

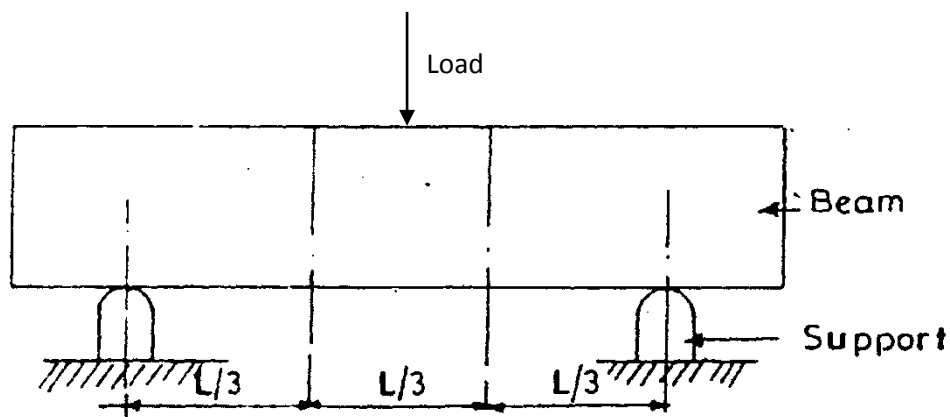


Fig. 3.1 Arrangement for Transverse Test on Timber

THEORETICAL BACKGROUND:

If force act on a piece of material in such a way that they tend to induce compressive stress over one part of cross-section and tensile stresses over the remaining part, the piece is said to be in bending. The common illustration of bending action is a beam acted upon by transverse loads.

The bending effect at any section is expressed as bending moment (M) which is the sum of the moment of all forces to the left (or to the right) of the section. The stresses induced by a bending moment may be termed as bending stress. For equilibrium, the resultant of the tensile forces (T) must always be equal to the resultant of compressive forces (C). The resultant of bending stress at any section from a couple that is equal in magnitude to the bending moment. The resultant of bending stresses at any section forms a couple that is equal in magnitude to the externally applied bending moment. The transverse shear (V) which is computed as the algebraic sum of all transverse forces to the left (or to the right) of section. Bending action in beams is often referred to as flexure.

In a cross-section of a beam, the line along which the bending stress is zero is called as the neutral axis. The surface containing the neutral axis of consecutive sections is called as the neutral surface. **Fig. 3.1** shows the arrangement of timber beam for transverse test.

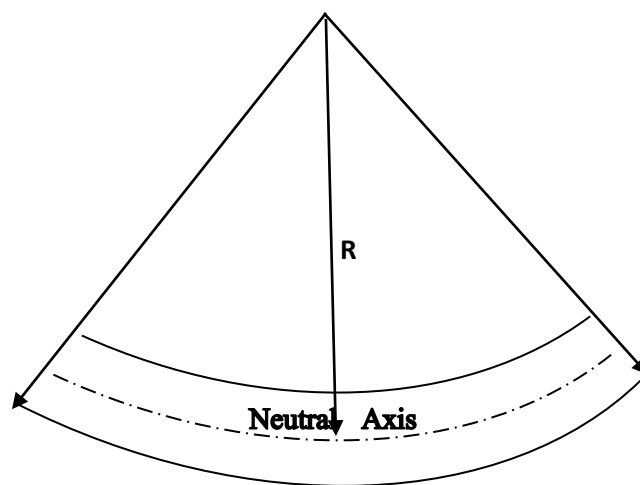


Fig. 4.2 Specimen after Bending

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

Fig. 3.2 shows the timber beam after bending. By summing the moment of the stress about neutral axis the resisting moments, within proportional limit, is found in terms of extreme fiber stress (f).

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

Where, σ = Extreme fiber bending stress, N / mm²
 y = Distance of the top fiber of the beam from the neutral axis, mm
 M = Bending moment, N mm
 I = Moment of inertia of cross-section of beam, mm⁴
 E = Young's Modulus of Elasticity of material, N / mm²
 R = Radius of curvature of the neutral layer, mm

The modulus of elasticity of the beam section, subjected to point load, is found from the deflection formulae,

$$\frac{WL^3}{48EI}$$

Where, W = Load applied at the center of the beam, kN
 L = Effective span, mm

The bending stress is calculated by formula, $\sigma = M y / I$ in N / mm².

PROCEDURE:

1. Measure the dimension of the timber beam specimen and place it on supports.
2. Note the span of simply supported beam.
3. Apply load gradually at the centre of the span.
4. Measure the deflection of the timber beam at desired interval of load with the help of scale fixed on the testing machine.
5. Plot the graph between W v/s δ .

APPLICATIONS:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

1. The bending test serves as a direct means of evaluating behavior under transverse loads, particularly for determining the limits of structural usefulness of beam of various shapes and sizes.
2. Flexural tests on beams are usually made to determine strength and stiffness in bending. Beam tests also offer a means of determining the resilience and toughness of materials in bending. The properties evaluated in the test can be used for design of beams and for selection of proper materials.

OBSERVATIONS:

Least Count of the Vernier caliper = _____

Sr. No.	Parameters	Specimen -1	Specimen -2
1.	Width of the beam (b), mm		
2.	Depth of the beam (d), mm		
3.	Effective span (L), mm		
4.	Distance of the top fiber of the beam from the neutral axis (y) = d / 2, mm		
5.	Ultimate Load W_{max} , N		
6.	Deflection at Ultimate load: δ_{max} , mm		

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CALCULATIONS:

Calculate Moment of inertia (I) and Modulus of Rupture (Stress at Failure) (σ)

1. Moment of Inertia is $I = b \cdot d^3 / 12$
2. Maximum Bending Moment is $M_{\max} = W_{\max} \cdot L / 4$
3. Modulus of Rupture is $\sigma_{\max} = M_{\max} \cdot y / I$
4. Young's Modulus, $E_1 = [\text{Slope} \cdot L^3] / [48 \cdot I]$
5. Average Young's Modulus, $E_{\text{avg}} =$

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CONCLUSIONS:

Grade Obtained:	Signature of Faculty:	Date:
------------------------	------------------------------	--------------

Date:

EXPERIMENT NO. 4
COMPRESSION TEST ON BRICKS & BLOCKS

OBJECTIVE:

To determine the compressive strength and to study the behavior of specimens of bricks and blocks when subjected to compressive load upto failure.

EQUIPMENT:

Universal Testing Machine.

APPARATUS:

Scale, Vernier caliper.

MATERIAL:

Brick & Cement Block

FIGURE:



Fig. 4.1 Universal Testing Machine

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

THEORETICAL BACKGROUND:

There are several materials used in engineering practice that are primarily meant to carry compressive load. Such material in compression member relevant mechanical properties is determined and load applied on a specimen up to failure or a predetermined load. **Fig. 4.1** shows the diagrams of the machine.

When a specimen is subjected to an axial load on end surfaces produces a crushing action. An internal resistance is setup against shortening of its length. This resistance is called compressive resistance. The compressive resistance is turned as compressive stress. Thus compressive stress is a ratio of compressive load (P) to the cross sectional area (A) resisting it.

PROCEDURE:

1. Place the specimen in position between the compression pads.
2. Switch on UTM.
3. Apply the load gradually and increase it steadily so as to avoid sudden shocks.
4. Note the ultimate load and breaking load.
5. Observe the failure of the specimen.

PRECAUTIONS:

1. Place the specimen at centre of compression pads.
2. The specimen should be free from defects.
3. Stop the UTM as soon as specimen fails.

LIMITATIONS:

1. It is difficult to ensure that the load applied on the specimen is truly concentric and axial.
2. There is always a tendency for bending stress to be set up, which may result into slipping of the specimen as the load increases.
3. Friction between the heads of the testing machine and the end surfaces of the specimen due to lateral expansion of the specimen may alter result considerably in compression to those obtained in absence of such lateral restraint.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

4. The relatively larger cross sectional areas of the specimen is required to obtain a proper degree of stability of the specimen.

APPLICATIONS:

1. The mechanical properties such as yield strength, modulus of toughness are determined by studying the behavior of specimen under load.
2. By experiment the strength of brick and cement block can be found and compare.
3. The Compressive strength can be found out.

OBSERVATIONS:

Specimen – 1: Brick

- (i) Size of Ordinary Brick = _____
- (ii) Size of Fly Ash Brick = _____

Specimen – 2: Cement Block

- (i) Size of Cement Block = _____

OBSERVATION TABLE:

Sr. No.	Observations	Specimen – 1	Specimen – 2	Specimen – 3	Specimen – 4
1.	Cross-sectional area (A), mm ²				
2.	Load at failure, N				
3.	Compressive strength, N/mm ²				
4.	Average Compressive strength, N/mm ²				

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CALCULATIONS:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CONCLUSIONS:

Comment on nature of fracture and the strength of the specimens.

Grade Obtained:	Signature of Faculty:	Date:
------------------------	------------------------------	--------------

Date:

EXPERIMENT NO. 5

TENSION TEST ON VARIOUS MATERIALS

OBJECTIVE:

To study the behavior of Steel under tensile load with the help of stress-strain curve on Universal Testing Machine.

EQUIPMENT:

Universal Testing Machine.

APPARATUS:

Vernier – Calliper and scale.

MATERIALS:

Mild steel and Cast iron.

RELATED I.S. CODES:

IS: 1608 – 1995: Mechanical testing of metals – Tensile strength.

IS: 1916 – 1979: Methods of tensile test for light metals and their alloys.

FIGURE:



Fig. 5.1 Universal Testing Machine

THEORETICAL BACKGROUND:

Tensile test is carried on Universal Testing Machine as shown in **Fig.5.1**. This machine is also used for testing materials subjected to compression, bending and shear tests.

The test consists of straining a test piece by tensile force, generally to fracture for the purpose of determining one or more mechanical properties viz. percentage elongation, percentage reduction in area, modulus of elasticity, yield stress, ultimate stress and breaking stress.

Specimen

The specimen is specially prepared as per prescribed standard guidelines, which are so framed as to induce failure within the length earmarked for observation – called the gauge length.

Test piece, the original gauge length of which is related to the original cross-sectional area (A_0) by the equation $L_0 = 5.65(\sqrt{A_0})$ is called proportional test piece. The portion in which the specimen is gripped is also highly stressed but the dimensions are so adjusted that these stresses do not affect failure within the gauge length. **Fig. 5.2** shows a typical specimen used for testing under tension.

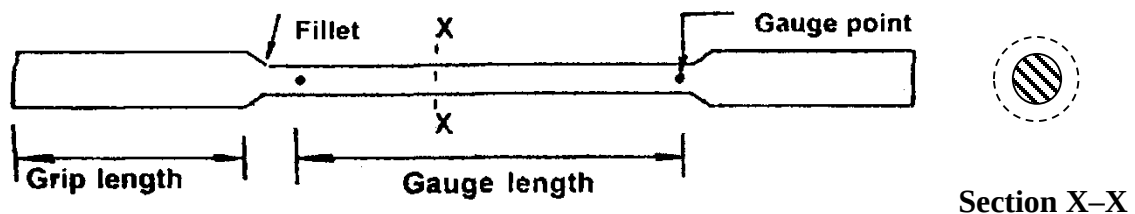


Fig. 2.2 Typical Specimen for Tensile Test

Metals used in engineering structures are mostly ductile, e.g. steel, aluminum, brass, copper etc, but some like cast iron are brittle. When specimens are tested under tensile (axial) load, both these types behave differently.

Ductile metals undergo comparatively large elongations, along the directions of loading, accompanied necessarily, of course, with contraction in the transverse direction. At a certain stress level, the specimen ceases to take any additional load but the elongation continues to a

certain level where after additional load is required to increase the elongation. At fracture, ultimately, the specimen breaks at or very near the narrowest section. The fracture is not along a plane right angle to cross-section; it is along an inclined plane and has appearance of a cup and cone shape. **Fig. 5.3** shows a typical fracture of a mild steel specimen. Obviously, therefore, fracture is not due to insufficient tensile resistance but due to insufficient shear resistance on inclined planes.



Fig. 5.3 Mild Steel Specimen after Fracture

The fracture of the surfaces of the specimen is depends on material. **Fig. 5.4** has been shown the different fracture surfaces for different materials.

Thus, the stages which a ductile material undergoes are yield-ultimate-breaking stress. The value of breaking if calculated with respect to is original area of the specimen, comes to be lesser than ultimate stress and is known as breaking stress whereas if breaking stress is calculated taking into consideration the instantaneous cross-sectional area, its value comes to be greater than ultimate stress and is known as nominal breaking stress or true stress. Since engineering design takes into consideration the design loads with respect to yield stress, true stress is generally not plotted in the stress-strain curve.

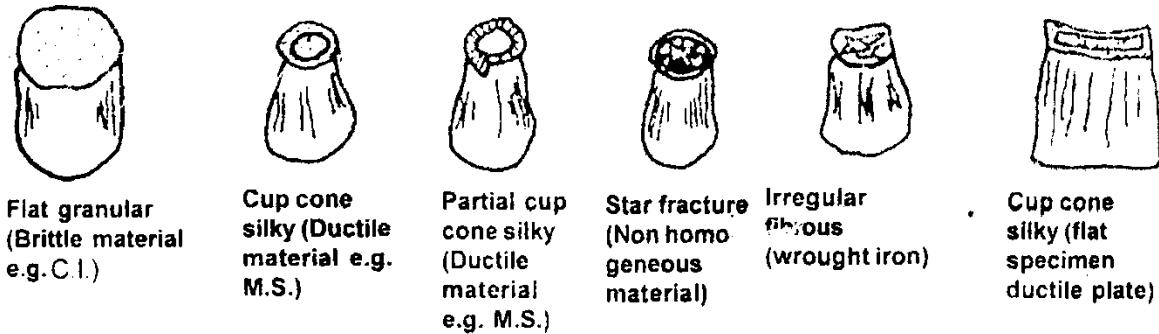


Fig. 5.4 Typical Tensile Fractures of Metals

PROPERTIES – DEFINITIONS:

Proportional limit:

Greatest stress that a material is capable of developing without deviation from straight line proportional between stress and strain. The ratio of stress to strain within this range is constant and is known as Modulus of Elasticity. Hooke's law defines this linear relationship.

Elastic limit:

Greatest stress that a material is capable of developing without a permanent set remaining upon complete release of stress.

Yield strength:

Stress above which material may be considered to be damaged and below which the damaging effect may be considered negligible.

Yield point:

Stress at which there occurs a marked increase in strain without increase in stress and without increase in stress.

Ultimate stress:

Stress corresponding to the maximum load applied.

Breaking stress:

Stress corresponding to the breaking load.

Toughness:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

Ability of the metal specimen to absorb energy during elastic and plastic deformation. The modulus of toughness is measured as **area under the entire stress-strain curve** and is the energy absorbed by the material of the specimen per unit volume up to fracture.

The standard values of Modulus of elasticity and elongation for different materials are given in **Table 5.1.**

Table 5.1 Standard Values of Modulus of Elasticity and Elongation for Different Materials

Material	Modulus of Elasticity 'E' (N/mm ²)	Elongation %
Duralumin	0.72×10^5	18
Cast Iron	0.96×10^5	08
Brass	1.02×10^5	40
Bronze	1.16×10^5	20
Timber	1.20×10^5	-
Monel Metal	1.80×10^5	20
Mild Steel	2.02×10^5	25
Nickel Crome Steel	2.06×10^5	12

The salient features of some of the important metals when subjected to this test are given in **Table 5.2.**

Table 5.2 Salient Features for Various Metals

Material	Elastic Curve	Yield Point	Stresses
Mild Steel, Structural Steel	Linear	Upper and lower are both distinctly visible	Yield-ultimate breaking
Aluminum, Copper and their alloys	Linear	Only one yield point	Yield-ultimate breaking
High yield strength deformed bar	Linear	Equivalent to proof stress	Yield-ultimate breaking
Cast Iron	Linear	-	Ultimate

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

Concrete	Non-linear	-	Ultimate
----------	------------	---	----------

PROCEDURE:

1. Measure the cross-sectional dimension of the given specimens and mark gauge points on the specimen with the help of hammer and punch or mark with chalk as per the instructions.
2. Adjust the capacity of UTM as per the anticipated value of load at failure of the test specimen.
3. Fix the specimen in the machine between the fixed cross-head and the upper cross-head, see that the specimen is symmetrical with respect to its longitudinal axis.
4. Apply the load gradually and increase it steadily so as to avoid sudden shocks.
5. Note the ultimate load and breaking load. Measure the final length and the reduced diameter of specimen after failure.
6. Observe the type of failure of the specimen.
7. Plot a graph of Stress versus Strain and find the value of the modulus of elasticity for elastic region.
8. Repeat the procedure for next specimen.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Faculty of Technology and Engineering**

Manubhai Shivabhai Patel Department of Civil Engineering

OBSERVATION TABLE – I:

Least count of vernier calipers: _____

Sr. No.	Parameters	Specimen -1	Specimen -2	Specimen -3	Specimen -4
Dimensions of the Specimen before Testing					
1.	Diameter (d_o), mm				
2.	Cross-sectional Area (A_o), mm ²				
3.	Gauge length ($L_o = 5.65\sqrt{A_o}$), mm				
Dimensions of the Specimen after Testing					
4.	Gauge length (L_f), mm				
5.	Diameter (d_1), mm				
	Diameter (d_2), mm				
	Final Diameter (d_f) =				

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Faculty of Technology and Engineering**

Manubhai Shivabhai Patel Department of Civil Engineering

	$\{(d_1 + d_2) / 2\}$, mm				
6.	Final Cross-sectional Area (A_f), mm ²				
Load at Critical Points					
7.	yield load (P_y), kN				
9.	Ultimate load (P_u), kN				
10.	Breaking load (P_f), kN				

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CALCULATIONS:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Faculty of Technology and Engineering**

Manubhai Shivabhai Patel Department of Civil Engineering

RESULT TABLE:

Sr. No.	Particulars	Formulae	Results for Specimen 1	Results for Specimen 2	Results for Specimen 3	Results for Specimen 4
1.	Yield stress, kN / mm	P_y / A_o				
3.	Ultimate stress , kN / mm ²	P_u / A_o				
4.	Nominal breaking stress, kN / mm ²	P_f / A_o				
5.	Actual breaking stress, kN / mm ²	P_f / A_f				
6.	Percentage reduction in area, %	$\{(A_o - A_f) / A_o\} * 100$				
7.	Percentage elongation, %	$\{(L_f - L_o) / L_o\} * 100$				
8.	Toughness	Area under stress strain curve				

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering

CONCLUSIONS:

Grade Obtained:	Signature of Faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Manubhai Shivabhai Patel Department of Civil Engineering